

## Arithmetic Mean (A.M):

$$\begin{aligned} &\rightarrow \text{Ungrouped} \rightarrow A.M = \frac{\sum_{i=1}^n x_i}{n} \\ &\rightarrow \text{grouped} \rightarrow A.M = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} \rightarrow \text{mid point} \end{aligned}$$

## Geometric Mean (G.M):

$$\rightarrow \text{Ungrouped} \rightarrow G.M = (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n}$$

## Harmonic Mean (H.M):

$$\rightarrow \text{Ungrouped} \rightarrow H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

Ex. Using the numbers 2, 4, 8, 5 and 7, provide a justification for the following relationships (i)  $A.M \gg G.M \gg H.M$

$$\Rightarrow A.M = \frac{\sum_{i=1}^n x_i}{n} = \frac{2+4+8+5+7}{5} = 5.2$$

$$G.M = (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n} = (2 \times 4 \times 8 \times 5 \times 7)^{1/5} = 4.677$$

$$H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{5}{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{5} + \frac{1}{7}} = 4.108$$

$\therefore A.M \gg G.M \gg H.M$  (shown) (Am)

Mean Deviation: The average deviation is the arithmetic mean of the absolute values of the deviations from the mean/median/mode

For ungrouped,

$$\text{Mean, M.D.}(\bar{x}) = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$$

$$\text{Median, M.D.}(M_e) = \frac{\sum_{i=1}^n |x_i - M_e|}{n}$$

$$\text{Mode, M.D.}(M_o) = \frac{\sum_{i=1}^n |x_i - M_o|}{n}$$

For grouped,

$$\text{Mean, M.D.}(\bar{x}) = \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{n} \rightarrow [n = \sum f_i]$$

$$\text{Median, M.D.}(M_e) = \frac{\sum_{i=1}^n f_i |x_i - M_e|}{n} \rightarrow [n = \sum f_i]$$

$$\text{Mode, M.D.}(M_o) = \frac{\sum_{i=1}^n f_i |x_i - M_o|}{n} \rightarrow [n = \sum f_i]$$

Variance: The variance is the arithmetic mean of the squared deviations from the mean. It is denoted by  $\sigma^2$

$$\therefore \sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Standard deviation: The positive square root of the variance is called standard deviation. It is denoted by  $\sigma$ .

$$\therefore \sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

For Ungrouped:

Standard deviation is

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$= \frac{\sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2)}{n}$$

$$= \frac{\sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + n\bar{x}^2}{n}$$

$$= \frac{\sum_{i=1}^n x_i^2 - 2\bar{x}n\bar{x} + n\bar{x}^2}{n}$$

$$= \frac{\sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2}{n}$$

$$= \frac{\sum_{i=1}^n x_i^2 - n\bar{x}^2}{n}$$

$$= \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2$$

$$= \frac{\sum_{i=1}^n x_i^2}{n} - \left( \frac{\sum_{i=1}^n x_i}{n} \right)^2$$

$$\therefore \sigma = \sqrt{\frac{\sum_{i=1}^n x_i^2}{n} - \left( \frac{\sum_{i=1}^n x_i}{n} \right)^2}$$

$$\left[ \begin{aligned} \bar{x} &= \frac{\sum_{i=1}^n x_i}{n} \\ &= n\bar{x} \\ &= \sum_{i=1}^n x_i \end{aligned} \right]$$



Variance:

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \text{ (for Ungrouped) } \text{ or } \sigma^2 = \frac{\sum_{i=1}^n x_i^2}{n} - \left( \frac{\sum_{i=1}^n x_i}{n} \right)^2$$

$$\sigma^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n} \text{ (for grouped) } \text{ or } \sigma^2 = \frac{\sum_{i=1}^n f_i x_i^2}{n} - \left( \frac{\sum_{i=1}^n f_i x_i}{n} \right)^2$$

Standard deviation:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}} \text{ (for Ungrouped) } \text{ or } \sigma = \sqrt{\frac{\sum_{i=1}^n x_i^2}{n} - \left( \frac{\sum_{i=1}^n x_i}{n} \right)^2}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}} \text{ (for grouped) } \text{ or } \sigma = \sqrt{\frac{\sum_{i=1}^n f_i x_i^2}{n} - \left( \frac{\sum_{i=1}^n f_i x_i}{n} \right)^2}$$

The lives (in hours) of 50 randomly selected flashlight batteries are:

| Class interval | 6.95-7.45 | 7.45-7.95 | 7.95-8.45 | 8.45-8.95 | 8.95-9.45 | 9.45-9.95 |
|----------------|-----------|-----------|-----------|-----------|-----------|-----------|
| frequency      | 2         | 10        | 18        | 11        | 5         | 4         |

using data verify that  $\sigma_x > MD_x$   $\left[ \bar{x} = \frac{\sum f_i x_i}{\sum f_i} \right]$

⇒

| Class interval | frequency $f_i$ | $x_i$ | $\bar{x}$ | $x_i - \bar{x}$ | $(x_i - \bar{x})^2$ | $f_i (x_i - \bar{x})$ | $f_i (x_i - \bar{x})^2$ |
|----------------|-----------------|-------|-----------|-----------------|---------------------|-----------------------|-------------------------|
| 6.95-7.45      | 2               | 7.2   | 8.5       | 1.24            | 1.537               | <del>2.48</del> 2.48  | 3.074                   |
| 7.45-7.95      | 10              | 7.8   | 8.5       | 0.64            | 0.409               | <del>6.4</del> 6.4    | 4.08                    |
| 7.95-8.45      | 18              | 8.2   | 8.5       | 0.24            | 0.057               | 4.32                  | 1.026                   |
| 8.45-8.95      | 11              | 8.8   | 8.5       | 0.36            | 0.129               | 3.96                  | 1.414                   |
| 8.95-9.45      | 5               | 9.2   | 8.5       | 0.76            | 0.577               | 3.8                   | 2.885                   |
| 9.45-9.95      | 4               | 9.8   | 8.5       | 1.36            | 1.849               | 5.44                  | 7.396                   |

$$h = 50$$

$$26.4$$

$$19.165$$

$$\therefore \sigma_x = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n}} = \sqrt{\frac{19.165}{50}} = 0.68$$

$$\therefore M.D(\bar{x}) = \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{n} = \frac{26.4}{50} = 0.528$$

$$\therefore \sigma_x > M.D(\bar{x}) \text{ (shown) } \text{ (Ans)}$$

Q. Using the numbers 2, 4, 8, 5 and 7, provide a justification for the following relationships: ii)  $\sigma_x > MD_x$

$\Rightarrow$  Here Mean derivation of 2, 4, 8, 5, 7

$$\bar{x} = \frac{2+4+8+5+7}{5} = 5.2 \quad \therefore n=5$$

| $x_i$             | $ x_i - \bar{x} $              | $(x_i - \bar{x})^2$                |
|-------------------|--------------------------------|------------------------------------|
| 2                 | $ 2 - 5.2  = 3.2$              | 10.24                              |
| 4                 | 1.2                            | 1.44                               |
| 8                 | 2.8                            | 7.84                               |
| 5                 | 0.2                            | 0.04                               |
| 7                 | 1.8                            | 3.24                               |
| $\Sigma x_i = 26$ | $\Sigma  x_i - \bar{x}  = 9.2$ | $\Sigma (x_i - \bar{x})^2 = 22.96$ |

$$\therefore \sigma_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}} = \sqrt{\frac{22.96}{5}} = 2.14$$

$$\therefore MD_x = \frac{\sum_{i=1}^n (x_i - \bar{x})}{n} = \frac{9.2}{5} = 1.84$$

$$\therefore \sigma_x > MD_x \text{ (shown) } \text{ (Ans)}$$

Ex. Show that the standard deviation of first  $n$  natural number is  $\sqrt{\frac{n^2-1}{12}}$

⇒ Assume  $x_i = 1, 2, 3, \dots, n$

$$\begin{aligned}\bar{x} &= \frac{\sum x_i}{n} \\ &= \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \\ &= \frac{1+2+3+\dots+n}{n} = \frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}\end{aligned}$$

$$\begin{aligned}\text{Standard variance, } \sigma^2 &= \frac{\sum x_i^2}{n} - (\bar{x})^2 \\ &= \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n} - \left(\frac{n+1}{2}\right)^2 \\ &= \frac{\frac{n(n+1)(2n+1)}{6}}{n} - \left(\frac{n+1}{2}\right)^2 \\ &= \frac{n+1}{2} \left[ \frac{2n+1}{3} - \frac{n+1}{2} \right] \\ &= \frac{n+1}{2} \left[ \frac{4n+2-3n-3}{6} \right] \\ &= \frac{n+1}{2} \times \frac{n-1}{6} = \frac{n^2-1}{12}\end{aligned}$$

∴ Standard deviation,  $\sigma = \sqrt{\frac{n^2-1}{12}}$  (Shoceed) (Am)



11. Find the arithmetic mean of the first  $n$  natural numbers whose frequencies are equal to the corresponding numbers.

⇒ The first ' $n$ ' natural numbers are  $1, 2, 3, \dots, n$  and their frequencies corresponding numbers are  $1, 2, 3, \dots, n$

$$\therefore \text{Arithmetic Mean, } \bar{A.M.} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

$$= \frac{1 \times 1 + 2 \times 2 + 3 \times 3 + \dots + n \times n}{1 + 2 + 3 + \dots + n}$$

$$= \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{1 + 2 + 3 + \dots + n}$$

$$= \frac{n(n+1)(2n+1)}{2}$$

$$= \frac{1}{3} (2n+1) \quad \text{'' (Ans)}$$

Co-efficient of Variation: The co-efficient of variation is the ratio between the standard deviation of a sample and its mean.

$$C.V = \frac{\sigma}{\bar{x}}$$

The co-efficient of variation is usually expressed in percentages:  $C.V = \frac{\sigma}{\bar{x}} \times 100$ .

Standard error of the mean (SEM): The standard error of the mean (SEM) measures how much discrepancy is likely in a sample's mean compared with the population mean.

$$SEM = \frac{\sigma}{\sqrt{n}}$$

Ex. Following is the frequency distribution of life length in hours of 20 electric bulbs:

| Life length | 8.5-13.5 | 13.5-18.5 | 18.5-23.5 | 23.5-28.5 | 28.5-33.5 |
|-------------|----------|-----------|-----------|-----------|-----------|
| frequency   | 3        | 5         | 7         | 4         | 1         |

Calculate the coefficient of variation (CV) and the standard error of mean (SEM) using the above data and comment on your results.

$$\bar{x} = 19.75$$

⇒

| Life length | frequency | $x_i$           | $\sum f_i x_i$       | $ x_i - \bar{x} $              | $(x_i - \bar{x})^2$               | $f_i  x_i - \bar{x} $              |
|-------------|-----------|-----------------|----------------------|--------------------------------|-----------------------------------|------------------------------------|
| 8.5-13.5    | 3         | 11              | 33                   | 8.75                           | 76.56                             | 229.68                             |
| 13.5-18.5   | 5         | 16              | 80                   | 3.75                           | 14.06                             | 70.3                               |
| 18.5-23.5   | 7         | 21              | 147                  | 1.25                           | 1.56                              | 10.92                              |
| 23.5-28.5   | 4         | 26              | 104                  | 6.25                           | 39.06                             | 156.24                             |
| 28.5-33.5   | 1         | 31              | 31                   | 11.25                          | 126.56                            | 126.56                             |
|             |           | $\sum x_i = 89$ | $\sum f_i x_i = 395$ | $\sum (x_i - \bar{x}) = 31.25$ | $\sum (x_i - \bar{x})^2 = 258.32$ | $\sum f_i  x_i - \bar{x}  = 593.7$ |



$$\therefore C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{5.448}{19.75} \times 100$$

$$= 0.2758 \times 100 = 27.58\% \quad (Am)$$

$$\sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{n}} \quad [n = f_i = 20]$$

$$= \sqrt{\frac{598.7}{20}} = 5.448$$

$$\therefore SEM = \frac{\sigma}{\sqrt{f_i}} = \frac{5.448}{\sqrt{20}} \quad [f_i = n]$$

$$= 1.218 \quad (Am)$$

| 1-1 | 2-2 | 3-3 | 4-4 | 5-5 | 6-6 | 7-7 | 8-8 | 9-9 | 10-10 | 11-11 | 12-12 | 13-13 | 14-14 | 15-15 | 16-16 | 17-17 | 18-18 | 19-19 | 20-20 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10    | 11    | 12    | 13    | 14    | 15    | 16    | 17    | 18    | 19    | 20    |

Q. Suppose your midterm test score is 73 and your final exam test score is 85. Using weights of 30% for the midterm and 70% for the final exam. Compute the weighted average of your scores. If the minimum average for an A is 75, will you earn an A?

$$\Rightarrow \bar{X}_w = \frac{73 \times 0.3 + 85 \times 0.7}{0.3 + 0.7} = 81.4 \quad \left[ \bar{X}_w = \frac{w_1x_1 + w_2x_2 + \dots}{w_1 + w_2 + \dots} \right]$$

(Ans)

As my weighted average is 81.4 which is greater or equal to minimum average 75, so

I will earn an A. (Ans)

Q. Lives of two brands of private cars in a recent survey were found as follows:

| Life (No. of years) | 0-4 | 4-8 | 8-12 | 12-16 | 16-20 | 20-24 |
|---------------------|-----|-----|------|-------|-------|-------|
| Model A             | 5   | 9   | 11   | 7     | 6     | 2     |
| Model B             | 2   | 7   | 13   | 8     | 4     | 6     |

P.T.O

(i) Which of the two brands shows more uniformity?

(ii) Suppose Mrs. Arif wants to buy a new car, which brand of car will be prefer?

⇒ i) ~~Calculate~~ Table for co-efficient variation:

| Model A ( $x_i$ ) | Model B ( $y_i$ ) | $x_i^2$            | $y_i^2$            |
|-------------------|-------------------|--------------------|--------------------|
| 5                 | 2                 | 25                 | 4                  |
| 9                 | 7                 | 81                 | 49                 |
| 11                | 13                | 121                | 169                |
| 7                 | 8                 | 49                 | 64                 |
| 6                 | 4                 | 36                 | 16                 |
| 2                 | 6                 | 4                  | 36                 |
| $\sum x_i = 40$   | $\sum y_i = 40$   | $\sum x_i^2 = 316$ | $\sum y_i^2 = 338$ |

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n} = \frac{40}{6} = 6.67$$

$$\bar{Y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{40}{6} = 6.67$$

$$\begin{aligned}\therefore CV(A) &= \frac{\sigma}{\bar{X}} \times 100 = \frac{\sqrt{\frac{\sum x_i^2}{n} - (\bar{X})^2}}{\bar{X}} \times 100 \\ &= \frac{\sqrt{\frac{316}{6} - (6.67)^2}}{6.67} \times 100 \\ &= 42.87\% \end{aligned}$$



$$\begin{aligned}
 \therefore C.V(B) &= \frac{\sigma}{\bar{Y}} \times 100 \\
 &= \frac{\sqrt{\frac{\sum x_i^2}{n} - (\bar{Y})^2}}{\bar{Y}} \times 100 \\
 &= \frac{\sqrt{\frac{338}{6} - (6.67)^2}}{6.67} \times 100 \\
 &= 51.59\%
 \end{aligned}$$

As  $C.V(B) > C.V(A)$   $\therefore$  Model B Car is more uniformity.

ii) As  $C.V(B) > C.V(A)$  . So, Mr. Arif wants to buy a car and he will prefer Brand is Model (B) Car.

P.F.O

Ex. Daily wages of workers of two factories A & B are as follows:

| Daily wages (T.K)           | 90-99 | 100-109 | 110-119 | 120-129 | 130-139 |
|-----------------------------|-------|---------|---------|---------|---------|
| No. of workers of factory A | 6     | 13      | 20      | 6       | 5       |
| No. of workers of factory B | 4     | 8       | 29      | 4       | 5       |

- Which factory has less variability wages
- Find combined mean and standard deviation of wages of the factories.

⇒

| Daily wages (T.K) | $f_i(A)$             | $f_i(B)$             | $x_i$ | $f_i(A) \cdot x_i$               | $f_i(B) \cdot x_i$               | $f(A) \cdot x_i^2$                   | $f(B) \cdot x_i^2$                   |
|-------------------|----------------------|----------------------|-------|----------------------------------|----------------------------------|--------------------------------------|--------------------------------------|
| 90-99             | 6                    | 4                    | 94.5  | 567                              | 378                              | 53581.5                              | 35721                                |
| 100-109           | 13                   | 8                    | 104.5 | 1358.5                           | 836                              | 141963.25                            | 87362                                |
| 110-119           | 20                   | 29                   | 114.5 | 2290                             | 3320.5                           | 262205                               | 380197.25                            |
| 120-129           | 6                    | 4                    | 124.5 | 747                              | 498                              | 93001.5                              | 62001                                |
| 130-139           | 5                    | 5                    | 134.5 | 672.5                            | 672.5                            | 90451.25                             | 90451.25                             |
|                   | $\Sigma f_i(A) = 50$ | $\Sigma f_i(B) = 50$ |       | $\Sigma f_i(A) \cdot x_i = 5635$ | $\Sigma f_i(B) \cdot x_i = 5705$ | $\Sigma f(A) \cdot x_i^2 = 641202.5$ | $\Sigma f(B) \cdot x_i^2 = 655732.5$ |

$$\bar{X}(A) = \frac{\Sigma f_i(A) x_i}{f_i(A)} = \frac{5635}{50} = 112.7$$

$$\bar{X}(B) = \frac{\Sigma f_i(B) x_i}{f_i(B)} = \frac{5705}{50} = 114.1$$

P.T.O

$$\begin{aligned}
 \therefore C.V(A) &= \frac{\sigma_A}{\bar{X}_{(A)}} \times 100 \\
 &= \frac{\sqrt{\frac{\sum f_i(A) x_i^2}{f_i(A)} - (\bar{X}_{(A)})^2}}{\bar{X}_{(A)}} \times 100 \\
 &= \frac{\sqrt{\frac{641202.5}{50} - (112.7)^2}}{112.7} \times 100 \\
 &= 9.83\%
 \end{aligned}$$

$$\begin{aligned}
 \therefore C.V(B) &= \frac{\sigma_B}{\bar{X}_{(B)}} \times 100 \\
 &= \frac{\sqrt{\frac{\sum f_i(B) x_i^2}{f_i(B)} - (\bar{X}_B)^2}}{\bar{X}_B} \times 100 \\
 &= \frac{\sqrt{\frac{655732.5}{50} - (114.1)^2}}{114.1} \times 100 \\
 &= 8.58\%
 \end{aligned}$$

As  $C.V(A) > C.V(B)$ , So factory B has less variability wages.

(Ans)



$$\text{ii) Combined Mean, } \bar{X}_{(AB)} = \frac{f_1(A) \times \bar{X}_{(A)} + f_1(B) \times \bar{X}_{(B)}}{f_1(A) + f_1(B)}$$

$$= \frac{50 \times 112.7 + 50 \times 114.1}{50 + 50}$$

$$= 113.4 \text{ " (Am)}$$

$$\text{Standard deviation, } \sigma_{(A)} = \sqrt{\frac{\sum_{i=1}^n f_i(A) x_i^2}{f_1(A)} - (\bar{X}_A)^2}$$

$$= \sqrt{\frac{641202.5}{50} - (112.7)^2}$$

$$= 11.07 \text{ " (Am)}$$

$$\text{Standard deviation, } \sigma_{(B)} = \sqrt{\frac{\sum_{i=1}^n f_i(B) x_i^2}{f_1(B)} - (\bar{X}_B)^2}$$

$$= \sqrt{\frac{655782.5}{50} - (114.1)^2}$$

$$= 9.78 \text{ " (Am)}$$

Measure of dispersion: The degree to which the numerical data tends to spread about an average value is called dispersion of data.

Types of measures dispersion:

There are two types of dispersion.

- 1) Absolute measure of dispersion
- 2) Relative measure of dispersion

Absolute measure of dispersion: These measures give us an idea about the amount of dispersion is a set of observations.

Different types of absolute measure of dispersion are:

- 1) Range
- 2) Semi-interquartile or Quartile deviation (Q.D)
- 3) Mean deviation (M.D)
- 4) ~~Variate~~ Variance and standard deviation

Relative measure of dispersion: Different types of relative measures are:

- 1) Co-efficient of Range
- 2) Co-efficient of quartile deviation
- 3) Co-efficient of mean deviation
- 4) Co-efficient of Variation (CV)

III. Difference between Mean deviation and standard deviation.



| Mean deviation                                                                 | Standard deviation                                                     |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 1) The average absolute deviation of each value from the mean of a data set,   | 1) The square root of the variance of a data set                       |
| 2) We use central points (mean, median, mode) to calculate the mean deviation  | 2) To calculate the standard deviation we only use the mean.           |
| 3) It lacks mathematical properties since only absolute values are considered. | 3) It is mathematically sound because algebraic signs are not ignored. |



Q. For  $n$  non-negative observations, show that,

$$CV \leq 100 \sqrt{n-1} \quad \text{or} \quad \bar{x} \sqrt{n-1} \geq \sigma$$

Ans) let,

$x_1, x_2, x_3, \dots, x_n$  be the set of observations

Then their arithmetic mean,  $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

We know that,

$$\sigma^2 = \frac{\sum_{i=1}^n x_i^2}{n} - (\bar{x})^2$$

$$\left( \sum_{i=1}^n x_i \right)^2 = \left( \sum_{i=1}^n x_i^2 \right) + \left( 2 \sum_{i < j} x_i x_j \right)$$

$$\left[ \because 2 \sum_{i < j} x_i x_j > 0 \right]$$

$$\therefore \left( \sum_{i=1}^n x_i \right)^2 > \left( \sum_{i=1}^n x_i^2 \right)$$

$$\text{or, } (n\bar{x})^2 > \left( \sum_{i=1}^n x_i^2 \right)$$

$$\text{or, } n^2 \bar{x}^2 > \left( \sum_{i=1}^n x_i^2 \right)$$

$$\text{or, } \frac{n^2 \bar{x}^2}{n} > \frac{\sum_{i=1}^n x_i^2}{n}$$

$$\text{or, } n\bar{x}^2 - \bar{x}^2 > \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2$$

$$\text{or, } \bar{x}^2 (n-1) > \sigma^2$$

$$\therefore \bar{x} \sqrt{n-1} > \sigma \quad \text{(shown)}$$

Again,

$$\bar{x} \sqrt{(n-1)} \geq \sigma$$

$$\text{or, } \sqrt{(n-1)} \geq \frac{\sigma}{\bar{x}}$$

$$\text{or, } 100 \sqrt{(n-1)} \geq \frac{\sigma}{\bar{x}} \times 100$$

$$\therefore 100 \sqrt{(n-1)} \geq CV \quad \text{,, (shown) (Ans)}$$